

GDYNIA OKSYWIE, Poland

WMO: 121420

Lat: 54.5857N

Lon: 18.5192E

Elev: 40

StdP: 100.85

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.7	-14.2	1.1	-10.1	-12.0	1.3	-7.8	16.1	4.0	14.2	4.0	3.4	150	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	26.2	20.0	24.6	19.3	23.1	18.6	21.3	24.7	20.2	23.3	19.3	22.1	4.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	14.8	23.4	19.0	13.9	22.1	18.1	13.1	21.2	62.0	24.8	58.2	23.4	55.0	22.1	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.1	8.7	-14.5	30.6	4.0	1.8	-17.4	31.9	-19.8	32.9	-22.0	33.9	-24.9	35.2		
(5)			-14.5	23.0	3.8	1.2	-17.3	23.9	-19.5	24.6	-21.7	25.3	-24.5	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-0.8	0.1	3.1	7.2	11.8	15.6	18.1	18.1	14.4	9.4	5.1	1.9	(6)
(7)		DBStd	7.44	4.64	4.11	3.88	3.56	3.48	2.88	2.46	2.44	2.50	3.27	2.99	4.19	(7)
(8)		HDD10.0	1394	334	277	216	98	18	1	0	0	1	51	147	251	(8)
(9)		HDD18.3	3573	592	511	473	334	204	92	34	34	120	276	396	508	(9)
(10)		CDD10.0	929	0	0	2	14	75	168	252	251	133	33	1	0	(10)
(11)		CDD18.3	68	0	0	0	0	3	9	28	26	2	0	0	0	(11)
(12)		CDH23.3	293	0	0	0	1	16	54	116	93	13	0	0	0	(12)
(13)		CDH26.7	41	0	0	0	0	2	9	17	12	2	0	0	0	(13)
(14)	Wind	WSAvg	4.1	4.9	4.7	4.6	3.9	3.6	3.8	3.5	3.4	3.7	4.1	4.3	4.7	(14)
(15)	Precipitation	PrecAvg	582	36	28	30	31	49	59	79	66	61	53	46	43	(15)
(16)		PrecMax	717	82	53	53	73	94	130	176	144	148	123	103	69	(16)
(17)		PrecMin	433	6	2	5	3	16	7	12	6	13	12	8	19	(17)
(18)		PrecStd	77	18	12	13	17	24	28	36	37	33	28	23	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	10.1	15.7	21.2	25.4	28.2	28.7	28.2	25.6	19.6	12.9	11.5	(19)
(20)			MCWB	8.8	7.8	9.9	12.9	18.0	20.3	21.0	20.6	17.9	15.1	10.3	9.5	(20)
(21)		2%	DB	8.1	8.3	12.1	17.7	22.3	24.7	26.1	25.6	21.8	17.4	11.5	9.6	(21)
(22)			MCWB	6.7	6.1	8.1	11.6	16.9	18.5	20.6	19.8	16.9	14.1	9.8	8.1	(22)
(23)		5%	DB	6.6	6.6	10.1	14.9	20.0	22.5	24.4	24.0	20.1	15.7	10.4	8.2	(23)
(24)			MCWB	5.3	4.9	6.9	10.2	15.7	17.7	19.7	19.2	16.2	12.9	9.1	6.8	(24)
(25)		10%	DB	5.3	5.4	8.3	12.9	17.8	20.7	22.9	22.7	18.8	14.2	9.3	7.0	(25)
(26)			MCWB	4.2	4.0	5.9	9.2	14.4	16.3	19.0	18.6	15.4	12.1	8.1	5.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	8.8	10.8	14.1	19.1	21.7	22.8	22.3	19.6	15.8	11.1	9.9	(27)
(28)			MCDB	10.4	9.4	13.8	19.1	23.5	26.7	27.0	25.8	23.3	18.2	12.1	11.2	(28)
(29)		2%	WB	6.8	6.2	8.9	12.1	17.5	19.5	21.6	21.0	17.8	14.5	10.1	8.4	(29)
(30)			MCDB	7.9	7.6	11.4	17.1	21.5	23.5	24.9	24.1	20.4	16.8	11.2	9.5	(30)
(31)		5%	WB	5.5	5.1	7.3	10.7	16.1	18.2	20.5	20.0	16.8	13.3	9.2	7.0	(31)
(32)			MCDB	6.4	6.6	9.4	14.2	19.4	21.6	23.5	23.1	19.3	15.3	10.2	7.9	(32)
(33)		10%	WB	4.2	4.0	6.1	9.5	14.6	17.0	19.5	19.1	15.9	12.3	8.2	5.8	(33)
(34)			MCDB	5.3	5.3	8.1	12.4	17.5	19.9	22.1	21.8	18.4	14.0	9.1	6.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	6.0	7.9	8.3	8.2	7.5	7.8	7.0	5.9	4.2	4.0	(35)
(36)			MCDBR	4.3	5.8	9.2	11.3	11.3	11.0	9.8	9.7	9.2	8.1	5.2	4.5	(36)
(37)			MCWBR	3.9	4.4	6.0	6.8	6.6	6.3	5.5	5.4	5.4	5.4	4.1	4.1	(37)
(38)		5% WB	MCDBR	4.2	5.7	8.1	10.6	10.4	9.9	8.8	8.7	8.4	7.4	5.0	4.4	(38)
(39)			MCWBR	4.0	4.6	5.7	6.7	6.3	6.2	5.4	5.4	5.5	5.3	4.1	4.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.339	0.365	0.408	0.399	0.395	0.425	0.418	0.378	0.360	0.344	0.310	(40)	
(41)		taud	2.288	2.293	2.289	2.226	2.323	2.366	2.298	2.321	2.410	2.429	2.327	2.287	(41)	
(42)		Ebn at Noon	623	735	803	814	847	853	816	798	786	710	576	539	(42)	
(43)		Edh at Noon	58	81	101	123	118	114	120	111	90	71	57	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.46	1.12	2.39	4.21	5.49	5.75	5.16	4.40	3.10	1.56	0.59	0.33	(44)	
(45)		RadStd	0.06	0.17	0.27	0.51	0.53	0.47	0.57	0.43	0.34	0.24	0.07	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-130	+67	N/A

Nomenclature: See separate page